

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure Equipment Megger OTS 80 AF Oil Test Set

Equipment Number: EQ-25
Serial Number: 102315737



1. PURPOSE

This standard operating procedure is intended to provide users the guide on the operation and maintenance of Megger OTS 80AF.

2. SCOPE

This standard operating procedure is employed in the GLTS Laboratory primarily for dielectric strength determination of high voltage insulating liquids, transformers, switchgear and other. Its automatic laboratory oil test sets perform accurate breakdown voltage tests. Molded test vessels give repeatable results with lock in precision electrode gap setting adjustment wheels. The transparent, shielded lid and large test chamber enable easy access to the test vessel.

3. TERMS AND DEFINITIONS

Dielectric breakdown voltage which is a measure of the oil's ability to withstand electric stress. A low breakdown voltage can indicate the presence of contaminants such as water or conducting particles.

4. RESPONSIBILITIES

Use, care and maintenance of the Megger OTS is directly assigned to the end user who is usually the laboratory chemist. This responsibility may be shared to other qualified user(s) upon the consent of the Sr. laboratory chemist. Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

5. SAFETY PRECAUTIONS

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- Use a proper electrical outlet of correct voltage and current handling capacity.
- Disconnect from the power supply prior to maintenance and servicing.
- Ensure that the operating switch operates properly.

6. PRINCIPLES OF OPERATION

- Application
 - a. The use of megger instrument is fully automatic. The operator has simply to prepare the test vessel, load it with sample oil, place it in test chamber, select the appropriate specification for the test and then start the test sequence.
 - b. The test set automatically carries out the sequence of test as defined by pre-selected national specification.
 - c. Oil testing specifications, for which the sets are pre-programmed are as ASTM D1816-04 & IEC 60156-1995-05.
 - d. A critical factor with respect to field of oil is its temperature, because samples from transformer taken shortly after switching off can be degrees hotter than equivalent room temperature laboratory tests performed. The megger instruments measure the oil temperature during the test so that temperature correction can be applied to results.

7. OPERATIONAL DETAILS

- Instrument Control Structure
 - a. Connect line power in the range 90 to 265V AC, 12V DC power circuit.
 - b. Turn on the instrument. The window contains six main tab icons:

Home contains summary information for each showing the selected automatic test standard.

Set up offers instrument setting of languages.

File access for USB or PC connection

Information offers details about instrument software configuration/ version.

Tools offers the ability to manage test standards including the ability to define custom tests, add or remove new standard and upgrade standard via USB or PC.

Language option represented by flag symbol.

- Navigation and Character Control
 - a. The keypad consists of arrow keys surrounding an option “OK” button for navigating the screens and selecting functions and options. These buttons fully control the instrument.
 - b. The up and down button arrows are used to navigate up and down functions listed on the left side of the screen. The “OK” key either scrolls through options associated with each functions or enter a lower-level window to perform a function.

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- c. Left and right arrow buttons navigate between tabs on the top-level window for optimum speed. Individual tab window saves the highlight function and return the same position on each screen.
 - d. A virtual query keypad that facilitates alphanumeric entry to specify asset ID, filename and custom test name. A character selected by navigating to it, current character shows with a blue background and pressing the “OK” button.
 - e. Numeric entry is via virtual keypad shown in the right screen.
- General Instruction of Breakdown Test
 - a. Select the “Home” tab window using the arrow keys and “OK” button. This window offers a summary of information required for a measurement. When the users select a particular standard only valid options are presented with respect to gap, electrodes and stirrer options.
 - b. Navigation is simply up and down arrow buttons move up and down the left side of the window whilst options for each line can be scrolled through using the “OK” key or information entered in case of Test ID.
 - c. Test in Progress: During the test the screen flashes red warning HV symbols to remind the user that a high voltage test is in progress. As safety measure, any key press will immediately halt the test. The electrode potential is displayed as it ramps up in oversized digits to emphasize the presence of high voltage. During the test there is a dwell time and possibly a stirring action. A countdown timer indicates period of dwell or stirring remaining.
 - d. Test Result: At the end of any test results are shown on the screen. The image shows the complete information presented to the users however the screen size naturally smaller so, a slider function enables the user to navigate through the result and print it if required. The result window facilities a direct print function at the bottom of the result window.

8. MAINTENANCE

- Vessel and Electrode Cleaning:
 - a. The oil test vessel and electrodes require no maintenance other than cleaning (Clean with jewelry solution and soft cloth on monthly basis). Electrodes can be unscrewed and replaced with another shape electrode.
 - b. Electrode should be examined regularly to ensure no pitting has occurred and the electrodes renewed where pitting is evident.

Note: The instrument includes a built-in electrode cleaning function which can be activated from the tool's menu. This function will perform 24 sequential breakdowns to remove deposited on electrode surfaces.

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- **Checking the voltage output** of megger OTS 80 AF by using Voltage check meter VCM100D/VCM80D.
 - The new OTS models are programmed with a test sequence suitable for using the VCM100D/VCM80D. This sequence causes the output of the oil test set to ramp up from 0 kV to the maximum voltage value, pausing at each 10 kV point for 10 seconds to allow the voltage output to be checked.
 - **Operation VCM100D/VCM80D:**
 - Simply replace the OTS test vessel with the VCM100D/VCM80D.
 - close the lid and navigate to, the VCM function on the tools tab. Select using the OK button.
 - After pressing the TEST button, the test will proceed automatically.
 - Read the VCM100D/VCM80D meter when each steady voltage is displayed up to the maximum instrument voltage.
 - **Cleaning VCM100D:** The VCM100D is not a user serviceable item but may require periodic cleaning using a clean cloth dampened with isopropanol.
- **Cleaning Megger Outer Surface:**
 - a. First disconnect from the line power and then wipe the instrument using a clean cloth dampened with isopropyl alcohol.
 - b. Wipe away any split oil in the chamber or on the outside of the test vessel with a lint free cloth.
 - c. When a lot of oil is split in the test chamber, the unit is designed with a drain facility at the rear. Simply unclip the clear tube and drain oil into a beaker or other suitable container.

REFERENCES

[1] Megger OTS100AF & Oil Test Set User Guide

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